Juggling Zeros in the Matrix (Example II)

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(second draft)

This is a sequel to the post *Skipping over Dimensions, Juggling Zeros in the Matrix* on my math blog *Shadows of Simplicity*.

Example II: Laguerre polynomials of order 1/2, OEIS A176230

Chebyshev, Hermite, Fibonacci and other polynomials are intimately related to the Pascal matrix. In fact, diagonal multiplication of ER (A086646) in Example I by the double factorials generates the coefficients of Laguerre polynomials of order -1/2, a subset of the coefficient matrix for the modified Hermite polynomials. These Laguerre polynomials expressed as differential operators satisfy Witt-Lie bracket relations.

Using the analysis and notation in Example I in the previous post, I'll just show the main equations.

The coefficient matrix for the modified Hermite polynomials starts with A099174 (aerated A034839)

$$AP = A \ddots P = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 3 \\ 0 \\ 15 \end{bmatrix} \ddots \begin{bmatrix} 1 & & & & & & \\ 1 & 1 & & & & & \\ 1 & 2 & 1 & & & & \\ 1 & 3 & 3 & 1 & & & \\ 1 & 4 & 6 & 4 & 1 & & \\ 1 & 5 & 10 & 10 & 5 & 1 & \\ 1 & 6 & 15 & 20 & 15 & 6 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & & & & & & \\ 0 & 1 & & & & & \\ 1 & 0 & 1 & & & & \\ 0 & 3 & 0 & 1 & & & \\ 3 & 0 & 6 & 0 & 1 & & \\ 0 & 15 & 0 & 10 & 0 & 1 & \\ 15 & 0 & 45 & 0 & 15 & 0 & 1 \end{bmatrix}$$

where the vector A is A123023, the aerated double factorials A00147, with e.g.f. $e^{x^2/2}$, so the e.g.f. for AP is

$$e^{x^2/2}e^{xt}.$$

Per the analysis in Example I, the inverse matrix AP^{-1} is generated by the vector B whose e.g.f. is $e^{-x^2/2}$, so it is a signed version of AP, with the e.g.f. $e^{-x^2/2}e^{xt}$.

The deaerated, even-row submatrix of AP is comprised of the coefficients of the normalized Laguerre polynomials of order -1/2, A176230, and begins with

$$ER = \begin{bmatrix} 1 & & & & \\ 1 & 1 & & & \\ 3 & 6 & 1 & & \\ 15 & 45 & 15 & 1 & \\ 105 & 420 & 210 & 28 & 1 \end{bmatrix}$$

with the even power generating function

$$e^{x^2/2}\cosh(x\sqrt{t}).$$

Its inverse is comprised of the dearated odd rows of AP^{-1}

$$ER^{-1} = \begin{bmatrix} 1 & & & & \\ -1 & 1 & & & \\ 3 & -6 & 1 & & \\ -15 & 45 & -15 & 1 & \\ 105 & -420 & 210 & -28 & 1 \end{bmatrix}$$

with the even power generating function

$$e^{-x^2/2}\cosh(x\sqrt{t}).$$

For the odd rows, we have the coefficients of the normalized Laguerre polynomials of order 1/2, A130757, beginning with

$$OR = \begin{bmatrix} 1 & & & & \\ 3 & 1 & & & \\ 15 & 10 & 1 & & \\ 105 & 105 & 21 & 1 & \\ 940 & 1260 & 378 & 36 & 1 \end{bmatrix}$$

with the odd power generating function

$e^{x^2/2}\sinh(x\sqrt{t})/\sqrt{t}$.

Its inverse begins with

$$OR^{-1} = \begin{bmatrix} 1 & & & & \\ -3 & 1 & & & \\ 15 & -10 & 1 & & \\ -105 & 105 & -21 & 1 & \\ 940 & -1260 & 378 & -36 & 1 \end{bmatrix}$$

with the odd power generating function

$e^{-x^2/2}\sinh(x\sqrt{t})/\sqrt{t}$.

The e.g.f. for the deaerated AP A034839 is the sum of the e.g.f.s of ER and OR

$e^{t^2/2}[\cosh(x\sqrt{t}) + \sinh(x\sqrt{t})/\sqrt{t})]$.

A lowering op for the rows of ER and ER^{-1}, gives, as in the first example,

$(2\sqrt{x}D_x)^2 R_n(x) = (2n)(2n-1)R_{n-1}(x)$,

and a lowering op for the rows of OR and OR^{-1}, gives

$\frac{1}{\sqrt{x}}(2\sqrt{x}D_x)^2\sqrt{x}\ R_n(x) = (2n+1)(2n)R_{n-1}(x)$.

The raising ops for the Hermite polynomials AP and signed AP, its inverse, AP^{-1}, are, respectively,

$$ROP_{\pm} = \mathfrak{R}_{\pm} = D_t \ \ln[e^{\pm t^2/2} e^{xt}] \mid_{t=D_x} = x \pm D_x,$$

with

$$\mathfrak{R}_{+}\mathfrak{R}_{-} = (x + D_x)(x - D_x) = x^2 - xD_x + 1 + xD_x - D_x^2 = 1 + x^2 - D_x^2,$$

and

$$\mathfrak{R}_{-}\mathfrak{R}_{+} = (x - D_x)(x + D_x) = x^2 - D^2 + [x, D] = -1 + x^2 - D_x^2,$$

so

$$\frac{\mathfrak{R}_{+}\mathfrak{R}_{-} + \mathfrak{R}_{-}\mathfrak{R}_{+}}{2} = x^2 - D^2,$$

the heat, or Schrodinger (with D replaced by iD), operator.

Note also that given the e.g.f. for the moments of an Appell polynomial sequence, such as AP and AP^{-1}, the associated differential op formed from the e.g.f. of its fundamental moment sequence generates the Appell polynomials as, for example,

$$e^{D_z^2/2} z^n = AP_n(z)$$

and

$$e^{-D_z^2/2} z^n = AP_n^{-1}(z),$$

and then, of course, we have the umbral compositional inverse result

$$z^n$$

$$= e^{D_z^2/2} e^{-D_z^2/2} z^n = e^{D_z^2/2} AP_n^{-1}(z) = AP_n^{-1}(AP.(z))$$

$$= e^{-D_z^2/2} e^{D_z^2/2} z^n = e^{-D_z^2/2} AP_n(z) = AP_n(AP^{-1}.(z)),$$

a particular example of the more general Appell relations

$$z^n = A(D_z) \frac{1}{A(D_z)} z^n = A(D_z) A_n^{-1}(z) = A_n^{-1}(A.(z))$$

$= \frac{1}{A(D_z)} A(D_z) z^n = A^{-1}(D_z) A_n(z) = A_n(A^{-1}.(z))$.

Connections to Laguerre polynomials, infinigens, and classic differential ops

The row polynomials of the two matrices ER and OR are intimately related to the Laguerre polynomials of order 1/2 and 1/2, respectively, which, in turn, are related to classic differential operators and infinitesimal generators. Following some results that I present in my pdf "Infinitesimal Generators, the Pascal Pyramid, and the Witt and Virasoro Algebras" in my math WordPress blog Shadows of Simplicity in the post "Infinigens, the Pascal Triangle, and the Witt and Virasoro algebras", bivariate row polynomials for the two matrices are

$$ER_n(x) = (2y)^n n! Lag_n^{-1/2}(-x/2y) = (2y)^n n! \sum_{k=0}^{n} \binom{n-\frac{1}{2}}{k-\frac{1}{2}} \frac{(\frac{x}{2y})^k}{k!}$$

with, e.g., $ER_3(x,y) = 15y^3 + 45y^2x + 15yx^2 + x^3$

and

$$OR_n(x) = (2y)^n n! Lag_n^{1/2}(-x/2y) = (2y)^n n! \sum_{k=0}^{n} \binom{n+\frac{1}{2}}{k+\frac{1}{2}} \frac{(\frac{x}{2y})^k}{k!}$$

with, e.g., $OR_3(x,y) = 105y^3 + 105y^2x + 21yx^2 + x^3$.

The e.g.f. for ER is

$\frac{e^{xt/(1-2t)}}{(1-2t)^{1/2}}$,

and for OR,

$\frac{e^{xt/(1-2t)}}{(1-2t)^{3/2}}$.

The lowering and raising ops, from the general Sheffer sequence formalism, for ER are given by

$\frac{D}{1+2D} ER_n(x) = (D - 2D^2 + 4D^3 - ...) ER_n(x) = nER_{n-1}(x)$

and

$$(1+2D+x(1+2D)^2)ER_n(x) = [\sqrt{x}(1+2D)]^2 ER_n(x) = ER_{n+1}(x).$$

Those for OR are given by

$$\frac{D}{1+2D}OR_n(x) = (D-2D^2+4D^3-...)OR_n(x) = nOR_{n-1}(x)$$

and

$$[3(1+2D)+x(1+2D)^2]OR_n(x) = [2(1+D)+(\sqrt{x}(1+2D))^2]OR_n(x) = OR_{n+1}(x)$$

Operational formulas for the normalized Laguerre polynomial revolve around the classic operator xDx, for which $(xDx)^n = x^n D^n x^n$.

ER is a generalized factorial matrix of the kind presented in A133314, with an infinitesimal matrix generator, or infingen, IER that is a matrix whose elements are all zeros except for the second diagonal, which are those of the second diagonal of ER, the hexagonal numbers A000384 (0,1,6,15,28,45,...), omitting the first 0; that is, the infinigen begins with

$$IER = \begin{bmatrix} 0 & & & & \\ 1 & 0 & & & \\ 0 & 6 & 0 & & \\ 0 & 0 & 15 & 0 & \\ 0 & 0 & 0 & 28 & 0 \end{bmatrix}$$

and

$$ER = \begin{bmatrix} 1 & & & & \\ 1 & 1 & & & \\ 3 & 6 & 1 & & \\ 15 & 45 & 15 & 1 & \\ 105 & 420 & 210 & 28 & 1 \end{bmatrix} = I + IER + \frac{IER^2}{2!} + \frac{IER^3}{3!} + \frac{IER^4}{4!}.$$

The infinigen IOR for OR has the nonzero subdiagonal with the elements (3,10,21,36,55,...), the second hexagonal numbers A014105 with the first 0 omitted. Interwoven, the diagonals of the two infingens form the sequence (1,,3,6,10,15,21,28,...), A000217, omitting the first zero, the components of the second column and diagonal of the Pascal matrix, so we've come naturally full circle back to our fundamental matrix. Pascal rules!

Full circle, yet we have the Riemann sphere to consider.

The generalized Laguerre polynomials can be characterized operationally through their Rodriguez formulas as well. For notational convenience and aesthetic value, define for any pair of operators $(\widehat{ST})^n = S^n T^n$. Our pair of normalized Laguerre polynomials have the op reps

$$n! Lag_n^{1/2}(-\widehat{xD}) = x^{-1/2} \widehat{Dx}^n x^{1/2}$$

$$= x^{-1/2} D^n x^n x^{1/2} = x^{-1/2} \tfrac{(xD+n+1/2)!}{(xD+1/2)!}$$

and

$$n! Lag_n^{-1/2}(-\widehat{xD}) = x^{1/2} \widehat{Dx}^n x^{-1/2}$$

$$= x^{1/2} D^n x^n x^{-1/2} = x^{1/2} \tfrac{(xD+n-1/2)!}{(xD-1/2)!},$$

where the last factorial relations are easily derived by action on x^s and are useful in deriving commutator relations and in suggesting the generalization of the generalized Laguerre polynomials to the Kummer confluent hypergeometric functions.

Again, this is related to the even subgroup of the modified Hermite polynomials by

$$n! Lag_n^{-1/2}(-x) = (-1)^n H_{2n}(i\sqrt{x})/2^{2n}.$$

A connection to a Witt-Lie algebra can be made using the op reps

$$L_n^{-1/2} = x^n n! Lag_n^{-1/2}(-\widehat{xD}) = x^n x^{1/2} \widehat{Dx}^n x^{-1/2}$$

$$= x^{1/2} x^n D^n x^n x^{-1/2} = x^{1/2} (xDx)^n x^{-1/2} = [x^{1/2}(xDx)x^{-1/2}]^n.$$

The op $(xDx)^n = x^n D^n x^n$ seems to have intrigued investigators of the calculus at least as early as Stokes and maybe Glaisher and can also be expressed as a conjugation of $z^2 D_z$, the infinigen for the special linear fractional transformation of $sl(C)$:

$$(zDz)^n = [z^{-1} z^2 Dz]^n = z^{-1}(z^2 D)^n z.$$

Then also from above

$$L_n^{-1/2} = x^n n! Lag_n^{-1/2}(-\widehat{xD})$$

$$= x^{1/2}(xDx)^n x^{-1/2} = [x^{1/2}(xDx)x^{-1/2}]^n$$

$$= x^{1/2}[x^{-1}(x^2D)^n x]x^{-1/2} = [x^{-1/2}(x^2D)x^{1/2}]^n = x^{-1/2}(x^2D)^n x^{1/2}$$

$$= x^n x^{1/2} \tfrac{(xD+n-1/2)!}{(xD-1/2)!}.$$

Similarly,

$$L_n^{1/2} = x^n n! Lag_n^{1/2}(-\widehat{xD})$$

$$= x^{-1/2}(xDx)^n x^{1/2} = x^{1/2}(x^2D)^n x^{-1/2}$$

$$= [x^{-1/2}(xDx)x^{1/2}]^n = [x^{1/2}(x^2D)x^{-1/2}]^n$$

$$= x^n x^{-1/2} \tfrac{(xD+n+1/2)!}{(xD+1/2)!}.$$

This association to the special linear fractional transformation can be anticipated from the generalization of the Laguerre polynomials to the Kummer confluent hypergeometric functions (most apparent from the last op equality) since they are known to be related to $sl(C)$.

Givental used xDx to construct a Witt-Lie algebra with the generator ops

$$l_n = -x^{-1/2}(xDx)^{n+1}x^{-1/2}$$

giving

$$l_n = (-x)L_{n+1}^{-1/2} = (-x)x^{n+1}(n+1)! Lag_{n+1}^{-1/2}(-\widehat{xD})$$

$$= (-x)[x^{-1/2}(x^2D)x^{1/2}]^{n+1} = (-x)x^{-1/2}(x^2D)^{n+1}x^{1/2}$$

$$= (-x)x^{n+1}x^{1/2} \tfrac{(xD+n+1/2)!}{(xD-1/2)!}.$$

As shown in my pdf, mentioned above, on the WItt and Virasoro algebras, l_n satisfies the defining bracket/commutator relation for a Witt-Lie algebra

$$[l_m, l_n] = l_m l_n - l_n l_m = (m-n) l_{m+n}.$$

Similarly, the Laguerre polynomials of degree 1/2 (or of any degree) can be massaged to give the Witt-Lie ops:

$$l_n = -x^{-1/2}(xDx)^{n+1}x^{-1/2} = [x^{-1/2}(xDx)x^{1/2}]^{n+1}(-1/x)$$

$$= [x^{n+1}(n+1)! Lag_{n+1}^{1/2}(-\widehat{xD})](-1/x).$$

Related Stuff

On xDx, the Laguerre operator, infinitesimal generator for numerous classic polynomial sequences:

a) A132440
b) A021009
c) A132792
d) A132710
e) A132681
f) A094587
g) A132159
h) A132013
i) A132014
j) A218234
k) A000522
l) "Infinitesimal Generators, the Pascal Pyramid, and the Witt and Virasoro Algebras", a pdf on my math WordPress blog Shadows of Simplicity in the post "Infinigens, the Pascal Triangle, and the Witt and Virasoro algebras"
m) "Mellin Interpolation of Differential Ops and Associated Infinigens and Appell Polynomials; The Ordered, Laguerre, and Scherk-Witt-Lie Operators" a post on my WordPress math blog

On x^2D, the infinitesimal generator for the special linear conformal / fractional / Moebius transformation and the operational generator for Lah polynomials (the Lageuerr polynomials of order -1 or 1, mod zero padding) :

a) A008297 (Nov. 3, 2016 formulas), another generalized factorial matrix
b) A218234
c) A097805
d) Lagrange a la Lah, a post at my WordPress math blog with two pdfs related to the op
e) Generators, Inversion, and Matrix, Binomial, and Integral Transforms, a post at my WordPress math blog with a pdf related to the op
f) A Class of Differential Operators and the Stirling Numbers, a post at my WordPress math blog with a pdf related to the op
g) "Mellin Interpolation of Differential Ops and Associated Infinigens and Appell Polynomials; The Ordered, Laguerre, and Scherk-Witt-Lie Operators" a post on my WordPress math blog
h) A094638
i) A133314
j) A111596
k) A105278
l) A130561

On the associated and generalized Laguerre polynomials and the Kummer confluent hypergeometric functions:

a) MathOverflow, Pocchammer Symbol of a Differential and Hypergeometric Polynomials, Question by Emilio Pisanty. Answer by Tom Copeland, 2012
b) A132681
c) Abramowitz and Stegun, Handbook of Mathematical Functions
d) Gradsheteyn and Ryzhik, Table of Integrals, Series, and Products

On Appell polynomial sequences:

a) A133314
b) A dozen or so entries on my WordPress blog Shadows of Simplicity and more in numerous entries in the OEIS (search "Copeland Appell" there).
c) Various articles by Feinsilver, Anshelevich, and Tempesta

On the finite operator calculus, umbral calculus, Sheffer polynomial sequences, special functions:

a) Math StackExchange, "What's umbral calculus about?" Question by Billy Rubina . Answer by Tom Copeland, 2014.
b) Mathemagical Forests, 2008 (and Addendum to Mathemagical Forests, 2010)

c) Numerous posts on my WordPress blog Shadows of Simplicity and entries in the OEIS (search “Copeland umbral” and “Copeland derivative operator” and “Copeland infinitesimal” there).
d) Wolfram MathWorld, Sheffer Sequence. (Algebraists often tend to cover up tracks in calculus and geometry, leaving naturally intuitive interpretations of their theory in the dust in their goal of general abstraction (and perhaps appearance of originality). Rota lamented about his role in this and the subsequent loss of rough-and-ready magic in umbral calculus once formally dressed, but he seemed chiefly interested in polynomials and the associated combinatorics so generally ignored the extension to more general power series and functions and such analytic methods as Mellin transform interpolation. To make easy sense of the MathWorld entry, which follows Roman, replace the formal t by a derivative op D_x in Eqn. 9 for a raising op for a general Sheffer sequence.)

On special functions such as the Hermite polynomials and their connections to Lie algebra/group theory:

a) “Classical Special Functions and Lie Groups” by Ryan Wasson and Robert Gilmore
b) “An Overview of the Relationship between Group Theory and Representation Theory to the Special Functions in Mathematical Physics” by Ryan Wasson
c) “Group Theory” by Robert Gilmore
d) “On the factorization method in quantum mechanics” by Oscar Rosas-Ortiz
e) “Representations and stochastics on groups of type-H” by Feinsilver and Kocik (see section on heat polynomials)